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Inventor(s)

KOSHIMIZU; Chishio

FREQUENCY CONTROL OF RADIO FREQUENCY POWER IN PLASMA PROCESSING

Abstract

In a disclosed plasma processing apparatus, a radio frequency power supply determines a first temporal change pattern of a source frequency of the radio frequency power which most suppresses a degree of reflection of the radio frequency power, while changing the source frequencies of first time points in a cycle of an electric bias and interpolating the source frequency in each of first division periods in the cycle divided by the first time points. The radio frequency power supply determines a second temporal change pattern of the source frequency which most suppresses the degree of reflection of the radio frequency power, while changing the source frequencies of second time points in the cycle from the first temporal change pattern or the derived pattern and interpolating the source frequency in each of second division periods in the cycle divided by the second time points.

Inventors: KOSHIMIZU; Chishio (Miyagi, JP)

Applicant: Tokyo Electron Limited (Tokyo, JP)

Family ID: 1000008669190

Assignee: Tokyo Electron Limited (Tokyo, JP)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of PCT Application No. PCT/JP2023/041648, filed on Nov. 20, 2023, which claims the benefit of priority from Japanese Patent Application No. 2022-191299, filed on Nov. 30, 2022. The entire contents of the above listed PCT and priority applications are incorporated herein by reference.

BACKGROUND

Field

[0002] The present disclosure relates to a plasma processing apparatus, a power supply system, and a frequency control method.

Description of the Related Art

[0003] A plasma processing apparatus is used in plasma processing to be performed on a substrate. The plasma processing apparatus generates plasma from a gas in a chamber by supplying a source radio frequency power. The plasma processing apparatus uses a bias radio frequency power to attract ions from the plasma generated in the chamber onto the substrate. Japanese Unexamined Patent Application Publication No. 2009-246091 discloses a plasma processing apparatus that modulates a power level and a frequency of a bias radio frequency power.

SUMMARY

[0004] A plasma processing apparatus is provided in an example embodiment. The plasma processing apparatus may include a chamber, a substrate support, a radio frequency power supply, and a bias power supply. The substrate support is disposed in the chamber. The radio frequency power supply is configured to supply a source radio frequency power to generate plasma from a gas in the chamber. The bias power supply is configured to supply an electric bias having a waveform cycle to the substrate support. The radio frequency power supply is configured to determine a first temporal change pattern of a source frequency of the source radio frequency power in the waveform cycle which most suppresses a degree of reflection of the source radio frequency power. The radio frequency power supply determines the first temporal change pattern by using respective source frequencies of the source radio frequency power at a plurality of first time points in the waveform cycle and an interpolated source frequency in each of a plurality of first division periods, while changing the respective source frequency at the plurality of first time points. The plurality of first division periods are divided by the plurality of first time points. The radio frequency power supply is configured to determine a second temporal change pattern of the source frequency in the waveform cycle which most suppresses a degree of reflection of the source radio frequency power. The radio frequency power supply determines the second temporal change pattern by using the respective source frequencies of the source radio frequency power at a plurality of second time points in the waveform cycle and an interpolated source frequency in each of a plurality of second division periods, while changing the respective source frequency at the plurality of second time points based on the first temporal change pattern or a temporal change pattern of the source

frequency generated from the first temporal change pattern. The plurality of second division periods are divided by the plurality of second time points. A number of the plurality of second division periods is greater than a number of the plurality of first division periods.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a diagram for illustrating a configuration example of a plasma processing system.

[0006] FIG. 2 is a diagram for illustrating a configuration example of a capacitively coupled plasma processing apparatus.

[0007] FIG. 3 is a diagram illustrating a configuration example of a power supply system in a plasma processing apparatus according to one example embodiment.

[0008] Each of FIG. 4A and FIG. 4B is a diagram illustrating an example of the waveform of an electric bias.

[0009] FIG. 5 is a diagram for describing an example of an optimization process of a temporal change pattern of a source frequency.

[0010] FIG. 6 is a diagram for describing an example of an optimization process of a temporal change pattern of a source frequency.

[0011] FIG. 7 is a diagram for describing an example of an optimization process of a temporal change pattern of a source frequency.

[0012] FIG. 8 is a diagram for describing an example of an optimization process of a temporal change pattern of a source frequency.

[0013] Each of FIG. 9A and FIG. 9B is a timing chart of an example of a source radio frequency power and an electric bias.

[0014] Each of FIG. 10A and FIG. 10B is a timing chart of an example of a source radio frequency power and an electric bias.

[0015] Each of FIG. 11A to FIG. 11C is a timing chart of an example of electric bias.

[0016] FIG. 12 is a flowchart illustrating a frequency control method according to one example embodiment.

[0017] FIG. 13 is a flowchart illustrating an example of operation STA of a frequency control method according to one example embodiment.

[0018] FIG. 14 is a flowchart illustrating an example of operation STA of a frequency control method according to one example embodiment.

DETAILED DESCRIPTION

[0019] Hereinafter, various example embodiments will be described in detail with reference to the drawings. In the drawings, the same or equivalent portions are denoted by the same reference symbols.

[0020] FIG. 1 illustrates an example configuration of a plasma processing system. In an embodiment, the plasma processing system includes a plasma processing apparatus 1 and a controller 2. The plasma processing system is an example substrate processing system, and the plasma processing apparatus 1 is an example substrate processing apparatus. The plasma processing apparatus 1 includes a plasma processing chamber 10, a substrate support 11, and a plasma generator 12. The plasma processing chamber 10 has a plasma processing space. The plasma processing chamber 10 further has at least one gas inlet for supplying at least one process gas into the plasma processing space and at least one gas outlet for exhausting gases from the plasma processing space. The gas inlet is connected to a gas supply 20 described below and the gas outlet is connected to a gas exhaust system 40 described below. The substrate support 11 is disposed in a plasma processing space and has a substrate support surface for supporting a substrate.

[0021] The plasma generator **12** is configured to generate a plasma from the at least one process gas supplied into the plasma processing space. The plasma formed in the plasma processing space may be, for example, a capacitively coupled plasma (CCP), an inductively coupled plasma (ICP), an electron-cyclotron-resonance (ECR) plasma, a helicon wave plasma (HWP), or a surface wave plasma (SWP).

[0022] The controller **2** processes computer executable instructions causing the plasma processing apparatus **1** to perform various operations described in this disclosure. The controller **2** may be configured to control individual components of the plasma processing apparatus **1** such that these components execute the various operations described herein. In an embodiment, the functions of the controller **2** may be partially or entirely incorporated into the plasma processing apparatus **1**. The controller **2** may include a processor **2a1**, a storage **2a2**, and a communication interface **2a3**. The controller **2** is implemented in, for example, a computer **2a**. The processor **2a1** may be configured to read a program from the storage **2a2**, and then perform various controlling operations by executing the program. This program may be preliminarily stored in the storage **2a2** or retrieved from any medium, as appropriate. The resulting program is stored in the storage **2a2**, and then the processor **2a1** reads to execute the program from the storage **2a2**. The medium may be of any type which can be accessed by the computer **2a** or may be a communication line connected to the communication interface **2a3**. The processor **2a1** may be a central processing unit (CPU) or a programmable logic device such as a FPGA (Field-Programmable Gate Array). The storage **2a2** may include a random access memory (RAM), a read only memory (ROM), a hard disk drive (HDD), a solid state drive (SSD), or any combination thereof. The communication interface **2a3** can communicate with the plasma processing apparatus **1** via a communication line, such as a local area network (LAN).

[0023] An example configuration of a capacitively coupled plasma processing apparatus, which is an example of the plasma processing apparatus **1**, will now be described. FIG. **2** illustrates the example configuration of the capacitively coupled plasma processing apparatus.

[0024] The capacitively coupled plasma processing apparatus **1** includes a plasma processing chamber **10**, a gas supply **20**, a power supply system **30**, and a gas exhaust system **40**. The plasma processing apparatus **1** further includes a substrate support **11** and a gas introduction unit. The gas introduction unit is configured to introduce at least one process gas into the plasma processing chamber **10**. The gas introduction unit includes a showerhead **13**. The substrate support **11** is disposed in a plasma processing chamber **10**. The showerhead **13** is disposed above the substrate support **11**. In an embodiment, the showerhead **13** functions as at least part of the ceiling of the plasma processing chamber **10**. The plasma processing chamber **10** has a plasma processing space **10s** that is defined by the showerhead **13**, the sidewall **10a** of the plasma processing chamber **10**, and the substrate support **11**. The plasma processing chamber **10** is grounded. The showerhead **13** and the substrate support **11** are electrically insulated from the housing of the plasma processing chamber **10**.

[0025] The substrate support **11** includes a body **111** and a ring assembly **112**. The body **111** has a central region **111a** for supporting a substrate **W** and an annular region **111b** for supporting the ring assembly **112**. An example of the substrate **W** is a wafer. The annular region **111b** of the body **111** surrounds the central region **111a** of the body **111** in plan view. The substrate **W** is disposed on the central region **111a** of the body **111**, and the ring assembly **112** is disposed on the annular region **111b** of the body **111** so as to surround the substrate **W** on the central region **111a** of the body **111**. Thus, the central region **111a** is also called a substrate support surface for supporting the substrate **W**, while the annular region **111b** is also called a ring support surface for supporting the ring assembly **112**.

[0026] In an embodiment, the body **111** includes a base **1110** and an electrostatic chuck **1111**. The base **1110** includes a conductive member. The electrostatic chuck **1111** is disposed on the base **1110**. The electrostatic chuck **1111** includes a ceramic member **1111a** and an electrostatic electrode

1111b disposed in the ceramic member **1111a**. The ceramic member **1111a** has the central region **111a**. In an embodiment, the ceramic member **1111a** also has the annular region **111b**. Any other member, such as an annular electrostatic chuck or an annular insulating member, surrounding the electrostatic chuck **1111** may have the annular region **111b**. In this case, the ring assembly **112** may be disposed on either the annular electrostatic chuck or the annular insulating member, or both the electrostatic chuck **1111** and the annular insulating member.

[0027] The ring assembly **112** includes one or more annular members. In an embodiment, the annular members include one or more edge rings and at least one cover ring. The edge ring is composed of a conductive or insulating material, whereas the cover ring is composed of an insulating material.

[0028] The substrate support **11** may also include a temperature adjusting module that is configured to adjust at least one of the electrostatic chuck **1111**, the ring assembly **112**, and the substrate to a target temperature. The temperature adjusting module may be a heater, a heat transfer medium, a flow passage **1110a**, or any combination thereof. A heat transfer fluid, such as brine or gas, flows into the flow passage **1110a**. In an embodiment, the flow passage **1110a** is formed in the base **1110**, one or more heaters are disposed in the ceramic member **1111a** of the electrostatic chuck **1111**. The substrate support **11** may further include a heat transfer gas supply configured to supply a heat transfer gas to a gap between the rear surface of the substrate **W** and the central region **111a**.

[0029] The showerhead **13** is configured to introduce at least one process gas from the gas supply **20** into the plasma processing space **10s**. The showerhead **13** has at least one gas inlet **13a**, at least one gas diffusing space **13b**, and a plurality of gas feeding ports **13c**. The process gas supplied to the gas inlet **13a** passes through the gas diffusing space **13b** and is then introduced into the plasma processing space **10s** from the gas feeding ports **13c**. The showerhead **13** further includes at least one upper electrode. The gas introduction unit may include one or more side gas injectors provided at one or more openings formed in the sidewall **10a**, in addition to the showerhead **13**.

[0030] The gas supply **20** may include at least one gas source **21** and at least one flow controller **22**. In an embodiment, the gas supply **20** is configured to supply at least one process gas from the corresponding gas source **21** through the corresponding flow controller **22** into the showerhead **13**. Each flow controller **22** may be, for example, a mass flow controller or a pressure-controlled flow controller. The gas supply **20** may include a flow modulation device that can modulate or pulse the

[0031] The gas exhaust system **40** may be connected to, for example, a gas outlet **10e** provided in the bottom wall of the plasma processing chamber **10**. The gas exhaust system **40** may include a pressure regulation valve and a vacuum pump. The pressure regulation valve enables the pressure in the plasma processing space **10s** to be adjusted. The vacuum pump may be a turbo-molecular pump, a dry pump, or a combination thereof.

[0032] Hereinafter, reference is made to FIG. 3 together with FIG. 2. FIG. 3 is a diagram illustrating a configuration example of a power supply system in the plasma processing apparatus according to one example embodiment. A power supply system **30** includes a radio frequency power supply **31** and a bias power supply **32**. The radio frequency power supply **31** configures a plasma generator **12** according to an embodiment. The radio frequency power supply **31** is configured to generate a source radio frequency power HF. The source radio frequency power HF has a source frequency $f_{sub.S}$. That is, the source radio frequency power HF has a sinusoidal waveform having a frequency of a source frequency $f_{sub.S}$. The source frequency $f_{sub.S}$ may be a frequency in a range of 10 MHz to 150 MHz.

[0033] The radio frequency power supply **31** is electrically connected to a radio frequency electrode via a matcher **33**, and is configured to supply the source radio frequency power HF to the radio frequency electrode. The radio frequency electrode may be disposed in a substrate support **11**. The radio frequency electrode may be at least one electrode provided in a conductive member of a base **1110** or a ceramic member **1111a**. Alternatively, the radio frequency electrode may be an upper electrode. When the source radio frequency power HF is supplied to the radio frequency

electrode, a plasma is generated from a gas in a chamber **10**.

[0034] The matcher **33** has variable impedance. The variable impedance of the matcher **33** is set to reduce the reflection from the load of the source radio frequency power HF. The matcher **33** may be controlled by, for example, a controller **2**.

[0035] In one embodiment, the radio frequency power supply **31** may include a signal generator **31g**, a D/A converter **31c**, and an amplifier **31a**. The signal generator **31g** generates a radio frequency signal having a source frequency $f_{sub.S}$. The signal generator **31g** may be configured by a programmable processor or a programmable logic device such as a Field-Programmable Gate Array (FPGA). The signal generator **31g** may be configured by a single programmable device **30p** together with a signal generator **32g** described below, or may be configured with a programmable device separate from the signal generator **32g**.

[0036] An output of the signal generator **31g** is connected to an input of the D/A converter **31c**. The D/A converter **31c** converts the radio frequency signal from the signal generator **31g** into an analog signal. An output of the D/A converter **31c** is connected to an input of the amplifier **31a**. The amplifier **31a** amplifies the analog signal from the D/A converter **31c** to generate the source radio frequency power HF. The amplification factor of the amplifier **31a** is designated from the controller **2** to the radio frequency power supply **31**. The radio frequency power supply **31** may not include the D/A converter **31c**. In this case, the output of the signal generator **31g** is connected to the input of the amplifier **31a**, and the amplifier **31a** amplifies the radio frequency signal from the signal generator **31g** to generate the source radio frequency power HF.

[0037] The bias power supply **32** is electrically coupled to the substrate support **11**. The bias power supply **32** is electrically connected to a bias electrode in the substrate support **11**, and is configured to supply an electric bias EB to the bias electrode. The bias electrode may be at least one electrode provided in the conductive member of the base **1110** or the ceramic member **1111a**. The bias electrode may be common to the radio frequency electrode. When the electric bias EB is supplied to the bias electrode, ions from the plasma are attracted to the substrate W.

[0038] Hereinafter, reference is made to FIG. 4A and FIG. 4B together with FIGS. 2 and 3. Each of FIG. 4A and FIG. 4B is a diagram illustrating an example of a waveform of the electric bias. The bias power supply **32** is configured to periodically apply the electric bias EB having a waveform cycle CY to the bias electrode. That is, the electric bias EB is applied to the bias electrode in each of a plurality of waveform cycles CY which are repetitions of the waveform cycle CY. The waveform cycle CY is defined by a bias frequency. The bias frequency is, for example, a frequency of 50 kHz or higher and 27 MHz or lower. The time length of the waveform cycle CY is the reciprocal of the bias frequency.

[0039] As illustrated in FIG. 4A, the electric bias EB may be bias radio frequency power LF having a bias frequency. That is, the electric bias EB may have a sinusoidal waveform of which a frequency is the bias frequency. In this case, the bias power supply **32** is electrically connected to the bias electrode via a matcher **34**. The variable impedance of the matcher **34** is set to reduce the reflection from the load of the bias radio frequency power LF.

[0040] Alternatively, as illustrated in FIG. 4B, the electric bias EB may include a voltage pulse VP. The voltage pulse VP is applied to the bias electrode in the waveform cycle CY. The voltage pulse VP is periodically applied to the bias electrode at a time interval equal to a length of the time length of the waveform cycle CY. The waveform of the voltage pulse VP may be a rectangular wave, a triangular wave, or any waveform. Polarity of the voltage of the voltage pulse VP is set so that a potential difference is generated between a substrate W and the plasma, and the ions from the plasma can be attracted onto the substrate W. The voltage pulse VP is applied to the bias electrode such that the waveform cycle CY includes a period in which the potential of the substrate W is a negative potential. The voltage pulse VP applied to the bias electrode may have a negative potential, may have a positive potential, or may have a potential that changes between a positive potential and a negative potential. The voltage pulse VP may be a negative voltage pulse or a

negative direct current voltage pulse. In a case where the electric bias EB is the voltage pulse VP, the plasma processing apparatus **1** may not include the matcher **34**.

[0041] In one embodiment, the bias power supply **32** may include a signal generator **32g**, a D/A converter **32c**, and an amplifier **32a** as shown in FIG. **3**. The signal generator **32g** periodically generates a bias signal having a designated waveform and a waveform cycle CY. The signal generator **32g** may be configured by a programmable logic device, such as a programmable processor or an FPGA.

[0042] An output of the signal generator **32g** is connected to an input of the D/A converter **32c**. The D/A converter **32c** converts the bias signal from the signal generator **32g** into an analog signal. An output of the D/A converter **32c** is connected to an input of the amplifier **32a**. The amplifier **32a** amplifies the analog signal from the D/A converter **32c** to generate the electric bias EB. The amplification factor of the amplifier **32a** is designated from the controller **2** to the bias power supply **32**. The bias power supply **32** may not include the D/A converter **32c**. In this case, the output of the signal generator **32g** is connected to the input of the amplifier **32a**, and the amplifier **32a** amplifies the bias signal from the signal generator **32g** to generate the electric bias EB. In a case where the electric bias EB is the voltage pulse VP, the bias power supply **32** may periodically switch the output voltage between the output voltage of the high voltage power supply and the ground potential or another potential to periodically generate the voltage pulse VP.

[0043] The radio frequency power supply **31** is configured to determine the optimal temporal change pattern of the source frequency $f_{\text{sub.S}}$ in the waveform cycle CY which most suppresses the degree of reflection of the source radio frequency power HF, by changing the source frequency $f_{\text{sub.S}}$ in a time series of the plurality of waveform cycles CY. Hereinafter, the process of determining the optimal temporal change pattern of the source frequency $f_{\text{sub.S}}$ is referred to as “optimization process”. In the optimization process, the radio frequency power supply **31** may set the source frequencies $f_{\text{sub.S}}$ at a plurality of discrete time points (phases) in each waveform cycle CY in the time series. Alternatively, the radio frequency power supply **31** may set the source frequency $f_{\text{sub.S}}$ continuously over time in each waveform cycle CY in the time series of the plurality of waveform cycles CY. The calculation of the optimization process can be performed in the signal generator **31g** of the radio frequency power supply **31**. The controller **2** may perform the calculation of the optimization process.

[0044] The plasma processing apparatus **1** may further include a sensor **35** and/or a sensor **36** to determine the degree of reflection of the source radio frequency power HF in the optimization process (see FIG. **2**). The sensor **35** is configured to measure a power level P_r of a reflected wave of the source radio frequency power HF from a load. The sensor **35** includes, for example, a directional coupler. The directional coupler may be provided between the radio frequency power supply **31** and the matcher **33**. The sensor **35** may be configured to further measure the power level P_f of the traveling wave of the source radio frequency power HF. The power level P_r of the reflected wave measured by the sensor **35** is notified to the radio frequency power supply **31**. In addition, the power level P_f of the traveling wave may be notified from the sensor **35** to the radio frequency power supply **31**.

[0045] The sensor **36** includes a voltage sensor and a current sensor. The sensor **36** is configured to measure a voltage $V_{\text{sub.S}}$ and a current $I_{\text{sub.S}}$ in a power feeding path that connects the radio frequency power supply **31** and the radio frequency electrode to each other. The source radio frequency power HF is supplied to the radio frequency electrode via the power feeding path. The sensor **36** may be provided between the radio frequency power supply **31** and the matcher **33**. The voltage $V_{\text{sub.S}}$ and the current $I_{\text{sub.S}}$ are notified to the radio frequency power supply **31**.

[0046] The radio frequency power supply **31** generates a representative value from the notified measurement value. The measurement value may be a power level P_r of the reflected wave acquired by the sensor **35**. The measurement value may be a value of a ratio (that is, a reflectivity) of a power level P_r of a reflected wave to an output power level of the source radio frequency

power HF. The measurement value may be a phase difference θ between the voltage V.sub.S and the current I.sub.S acquired by the sensor **36**. The measurement value may be the impedance Z.sub.L on the load side of the radio frequency power supply **31** in each of the plurality of phase periods SP. The impedance ZI, is determined from the voltage V.sub.S and the current I.sub.S acquired by the sensor **36**. The measurement value may be a reflection coefficient obtained from the impedance Z.sub.L and the output impedance of the radio frequency power supply **31**. The representative value represents a degree of reflection of the source radio frequency power HF in each waveform cycle CY. The representative value may be an average value or a maximum value of the measurement values in each waveform cycle CY. Alternatively, the representative value may be an average value or a maximum value of the measurement values in a partial period in which the source frequency f.sub.S is changed in each waveform cycle CY.

[0047] Hereinafter, the optimization process will be described in detail with reference to FIGS. 5 to 8. Each of FIGS. 5 to 8 is a diagram for describing an example of the optimization process of a temporal change pattern of a source frequency. The optimization process includes a first optimization process, . . . , and a J-th optimization process sequentially. J may be any integer which is equal to or greater than 2. In the examples shown in FIGS. 5 to 7, J is 3.

[0048] In the optimization process, the radio frequency power supply **31** searches for an optimal time change pattern in the time series of the plurality of waveform cycles CY. In the optimization process, the radio frequency power supply **31** uses an initial temporal pattern of the source frequency f.sub.S which is prepared in advance as the temporal pattern of the source frequency f.sub.S of the initial waveform cycle CY in the time series.

[0049] Next, the radio frequency power supply **31** performs the first optimization process. In the first optimization process, the radio frequency power supply **31** determines the first temporal change pattern of the source frequency f.sub.S in the waveform cycle CY which most suppresses the degree of reflection of the source radio frequency power HF.

[0050] In the first optimization process, the radio frequency power supply **31** uses a temporal change pattern of a source frequency f.sub.S including a source frequency f.sub.S of each of a plurality of first time points (phases) and an interpolated source frequency f.sub.S in each of a plurality of first division periods DP.sub.1, in each waveform cycle CY in the time series. The number of the plurality of first time points in each waveform cycle CY is equal to or greater than 2. In the example shown in FIG. 5, the plurality of first time points include a time point Tu to a time point T.sub.13. The waveform cycle CY is divided into the plurality of first division periods DP.sub.1 by the plurality of first time points. In the example shown in FIG. 5, the plurality of first division periods DP.sub.1 include a period between the time point T.sub.11 and the time point T.sub.12, a period between the time point T.sub.12 and the time point T.sub.13, and a period between the time point T.sub.13 and a final time point T.sub.E in the waveform cycle CY. The source frequency f.sub.S at the initial first time point T.sub.11 and the source frequency f.sub.S at the final time point T.sub.E in the waveform cycle CY are the same.

[0051] The plurality of first time points (phases) in each waveform cycle CY may be predetermined. Alternatively, the plurality of first time points may be set such that each of the plurality of first division periods DP.sub.1 has the first change amount as the change amount of the load impedance of the radio frequency power supply **31** with respect to the above-described initial temporal pattern. Alternatively, the plurality of first time points may be set such that each of the plurality of first division periods DP.sub.1 has the first change amount as the change amount in the source frequency with respect to the above-described initial temporal pattern. Alternatively, the plurality of first time points may be set such that a section in the waveform cycle CY is divided into a greater number of division periods as the load impedance of the radio frequency power supply **31** or the degree of reflection of the source radio frequency power HF is greater in the section.

Alternatively, in a case where the electric bias EB includes the voltage pulse VP, the plurality of first time points may be set such that a section in the waveform cycle CY is divided into a greater

number of division periods as the change amount in the voltage in the substrate support **11** according to the voltage pulse VP is greater in the section. Alternatively, in a case where the electric bias EB includes the voltage pulse VP, one of the plurality of first division periods DP.sub.1 may include a period in which the voltage pulse VP is supplied to the substrate support **11**.

[0052] In the first optimization process, the radio frequency power supply **31** changes the temporal change pattern of the source frequency f.sub.S of the waveform cycle CY.sub.m in the time series from a temporal change pattern of the source frequency f.sub.S in a preceding waveform cycle CY.sub.m-M1. Specifically, the radio frequency power supply **31** sequentially changes the respective source frequencies f.sub.S at the plurality of first time points in the time series of the plurality of waveform cycles CY. More specifically, the radio frequency power supply **31** sets the source frequency f.sub.S[m, T.sub.1n] at the first time point T_{in} in the waveform cycle CY.sub.m in the time series to a frequency having one change with respect to the source frequency f.sub.S[m-M1, T.sub.1n] at the first time point T_{in} in the preceding waveform cycle CY.sub.m-M1. The one change is a decrease or an increase in frequency. “m” is a natural number representing an order of the waveform cycle CY in the time series. “n” is a natural number representing an order of each of the plurality of first time points in the waveform cycle CY. “M1” is an integer which is equal to or greater than 1, and is, for example, 1. In addition, f.sub.S[m, T.sub.1n] represents a source frequency f.sub.S at a first time point T.sub.1n in the waveform cycle CY.sub.m.

[0053] The radio frequency power supply **31** interpolates the source frequency f.sub.S of each of the plurality of first division periods DP.sub.1 from the source frequency f.sub.S at the immediately before time point and the source frequency f.sub.S at the immediately after time point among the plurality of first time points and the time point T.sub.E. The radio frequency power supply **31** can use linear interpolation in the interpolation of the source frequency f.sub.S. Alternatively, the radio frequency power supply **31** may use interpolation by a high-order expression in the interpolation of the source frequency f.sub.S. In the example shown in FIG. 5, a situation in which the source frequency f.sub.S at the first time point T.sub.12 is changed is shown.

[0054] In the first optimization process, in a case where the degree of reflection of the source radio frequency power HF is reduced due to the one change, the radio frequency power supply **31** sets the source frequency f.sub.S[m+M1, T.sub.1n] in the subsequent waveform cycle CY.sub.m+M1 in the time series to a frequency having the one change with respect to the source frequency f.sub.S[m, T.sub.1n]. In a case where the degree of reflection of the source radio frequency power HF is increased due to the one change, the radio frequency power supply **31** sets the frequency f.sub.S[m+M1, T.sub.1n] in the subsequent waveform cycle CY.sub.m+M1 in the time series to a frequency having the other change with respect to the source frequency f.sub.S[m, T.sub.1n]. The other change is a decrease or an increase in frequency, which is a change opposite to the one change.

[0055] The radio frequency power supply **31** may sequentially optimize the respective source frequencies f.sub.S of the plurality of first time points to suppress the degree of reflection of the source radio frequency power HF to determine the first temporal change pattern in the first optimization process. Alternatively, the radio frequency power supply **31** may determine the first temporal change pattern by repeating the sequential adjustment (or change) of the respective source frequencies f.sub.S of the plurality of first time points to determine the first temporal change pattern. In any case, the order of the first time points, at which the source frequency f.sub.S is optimized or adjusted, among the plurality of first time points may be predetermined, or may be any order. For example, the source frequency f.sub.S may be optimized or adjusted sequentially from a time point at which it is empirically known that the change in the source frequency f.sub.S at that time point has a large influence on the degree of reflection among the plurality of first time points.

[0056] In the first optimization process, the radio frequency power supply **31** determines, as the

first temporal change pattern, the temporal change pattern of the source frequency $f_{\text{sub.S}}$ of the waveform cycle CY in a case where the degree of reflection of the source radio frequency power HF satisfies a first condition. The first condition is predetermined as a condition indicating that the degree of reflection of the source radio frequency power HF is sufficiently suppressed. For example, the first condition may be satisfied in a case where the degree of reflection of the source radio frequency power HF is included in a first allowable range. In FIG. 5, the initial temporal pattern is shown by a solid line, and the temporal change pattern used in the first optimization process is shown by a broken line.

[0057] The radio frequency power supply **31** performs the second optimization process after the first optimization process. In the second optimization process, the radio frequency power supply **31** determines the second temporal change pattern of the source frequency $f_{\text{sub.S}}$ in the waveform cycle CY which most suppresses the degree of reflection of the source radio frequency power HF, based on the first reference pattern. The first reference pattern is the first temporal change pattern or a first derived temporal pattern of a source frequency $f_{\text{sub.S}}$ generated from the first temporal change pattern.

[0058] In the second optimization process, the radio frequency power supply **31** uses a temporal change pattern of a source frequency $f_{\text{sub.S}}$ including the respective source frequencies $f_{\text{sub.S}}$ of a plurality of second time points (phases) and an interpolated source frequency $f_{\text{sub.S}}$ in each of a plurality of second division periods DP.sub.2, in each waveform cycle CY in the time series.

[0059] The number of the plurality of second time points in each waveform cycle CY is greater than the number of the plurality of first time points in each waveform cycle CY in the first optimization process. In the example shown in FIG. 6, the plurality of second time points include a time point T.sub.21 to a time point T.sub.27. The waveform cycle CY is divided into the plurality of second division periods DP.sub.2 by the plurality of second time points. The number of the plurality of second division periods DP.sub.2 in each waveform cycle CY is greater than the number of the plurality of first division periods DP.sub.1 in the waveform cycle CY in the first optimization process. In the example shown in FIG. 6, the plurality of second division periods DP.sub.2 include a period between the time point T.sub.21 and the time point T.sub.22, a period between the time point T.sub.22 and the time point T.sub.23, a period between the time point T.sub.23 and the time point T.sub.24, a period between the time point T.sub.24 and the time point T.sub.25, a period between the time point T.sub.25 and the time point T.sub.26, a period between the time point T.sub.26 and the time point T.sub.27, and a period between the time point T.sub.27 and the final time point T.sub.E in the waveform cycle CY. The source frequency $f_{\text{sub.S}}$ at the initial second time point T.sub.21 and the source frequency $f_{\text{sub.S}}$ at the final time point T.sub.E in the waveform cycle CY are the same.

[0060] A plurality of second time points (phases) in each waveform cycle CY may be predetermined. Alternatively, the plurality of second time points may be set such that each of the plurality of second division periods DP.sub.2 has the second change amount as the change amount of the load impedance of the radio frequency power supply **31** with respect to the first reference pattern described above. In this case, the second change amount is smaller than the first change amount. Alternatively, the plurality of second time points may be set such that each of the plurality of second division periods DP.sub.2 has the second change amount as the change amount in the source frequency with respect to the first reference pattern. Even in this case, the second change amount is smaller than the first change amount. Alternatively, the plurality of second time points may be set such that a section in the waveform cycle CY is divided into a greater number of division periods as the load impedance of the radio frequency power supply **31** or the degree of reflection of the source radio frequency power HF is greater in the section. Alternatively, in a case where the electric bias EB includes the voltage pulse VP, the plurality of second time points may be set such that a section in the waveform cycle CY is divided into a greater number of division periods as the change amount in the voltage in the substrate support **11** according to the voltage

pulse VP is greater in the section. The plurality of second time points may include a plurality of time points that are the same as the plurality of first time points in the waveform cycle CY.

[0061] In the second optimization process, the radio frequency power supply **31** changes the temporal change pattern of the source frequency $f_{\text{sub.S}}$ of the waveform cycle CY.sub.m in the time series from a temporal change pattern of the source frequency $f_{\text{sub.S}}$ in a preceding waveform cycle CY.sub.m-M2. In the initial stage of the second optimization process, the temporal change pattern of the source frequency $f_{\text{sub.S}}$ of the waveform cycle CY is obtained by changing the first reference pattern.

[0062] Specifically, the radio frequency power supply **31** sequentially changes the respective source frequencies $f_{\text{sub.S}}$ at the plurality of second time points in the time series of the plurality of waveform cycles CY. More specifically, in the second optimization process, the radio frequency power supply **31** sets the source frequency $f_{\text{sub.S}}[m, T_{\text{sub.2n}}]$ at the second time point $T_{\text{sub.2n}}$ in the waveform cycle CY.sub.m in the time series to a frequency having one change with respect to the source frequency $f_{\text{sub.S}}[m-M2, T_{\text{sub.2n}}]$ at the second time point $T_{\text{sub.2n}}$ in the preceding waveform cycle CY.sub.m-M2. The one change is a decrease or an increase in frequency. “m” is a natural number representing an order of the waveform cycle CY in the time series. “n” is a natural number representing an order of each of the plurality of second time points in the waveform cycle CY. “M2” is an integer which is equal to or greater than 1, and is, for example, 1. In addition, $f_{\text{sub.S}}[m, T_{\text{sub.2n}}]$ represents the source frequency $f_{\text{sub.S}}$ at the second time point $T_{\text{sub.2n}}$ in the waveform cycle CY.sub.m.

[0063] The radio frequency power supply **31** interpolates the respective source frequencies $f_{\text{sub.S}}$ of the plurality of second division periods DP.sub.2 from the source frequency $f_{\text{sub.S}}$ at the immediately before time point and the source frequency $f_{\text{sub.S}}$ at the immediately after time point among the plurality of second time points and the time point $T_{\text{sub.E}}$. The radio frequency power supply **31** can use linear interpolation in the interpolation of the source frequency $f_{\text{sub.S}}$. Alternatively, the radio frequency power supply **31** may use interpolation by a high-order expression in the interpolation of the source frequency $f_{\text{sub.S}}$.

[0064] In the second optimization process, in a case where the degree of reflection of the source radio frequency power HF is reduced due to the one change, the radio frequency power supply **31** sets the source frequency $f_{\text{sub.S}}[m+M2, T_{\text{sub.2n}}]$ in the subsequent waveform cycle CY.sub.m+M2 in the time series to a frequency having the one change with respect to the source frequency $f_{\text{sub.S}}[m, T_{\text{sub.2n}}]$. In a case where the degree of reflection of the source radio frequency power HF is increased due to the one change, the radio frequency power supply **31** sets the frequency $f_{\text{sub.S}}[m+M2, T_{\text{sub.2n}}]$ in the subsequent waveform cycle CY.sub.m+M2 in the time series to a frequency having the other change with respect to the source frequency $f_{\text{sub.S}}[m, T_{\text{sub.2n}}]$. The other change is a decrease or an increase in frequency, which is a change opposite to the one change.

[0065] The radio frequency power supply **31** may sequentially optimize the respective source frequencies $f_{\text{sub.S}}$ of the plurality of second time points to suppress the degree of reflection of the source radio frequency power HF to determine the second temporal change pattern in the second optimization process. Alternatively, the radio frequency power supply **31** may determine the second temporal change pattern by repeating the sequential adjustment (or change) of the respective source frequencies $f_{\text{sub.S}}$ at the plurality of second time points to determine the second temporal change pattern. In any case, the order of the plurality of second time points at which the source frequency $f_{\text{sub.S}}$ is optimized or adjusted may be predetermined among the plurality of second time points, or may be any order. For example, the source frequency $f_{\text{sub.S}}$ may be optimized or adjusted sequentially from a time point at which it is empirically known that the change in the source frequency $f_{\text{sub.S}}$ at that time point has a large influence on the degree of reflection among the plurality of second time points.

[0066] In the second optimization process, the radio frequency power supply **31** determines, as the

second temporal change pattern, the temporal change pattern of the source frequency $f_{\text{sub.S}}$ of the waveform cycle CY in a case where the degree of reflection of the source radio frequency power HF satisfies a second condition. The second condition is predetermined as a condition indicating that the degree of reflection of the source radio frequency power HF is sufficiently suppressed. For example, the second condition may be satisfied in a case where the degree of reflection of the source radio frequency power HF is included in a second allowable range. In FIG. 6, the first reference pattern is shown by a solid line, and the temporal change pattern used in the second optimization process is shown by a broken line.

[0067] The radio frequency power supply 31 may further perform a third optimization process after the second optimization process. Further, the radio frequency power supply 31 may further perform a fourth optimization process after the third optimization process. Hereinafter, the j -th optimization process which represents each of the second optimization process to the J -th optimization process will be described using a reference numeral “ j ” as a reference numeral indicating an order of each optimization process.

[0068] In the j -th optimization process, the radio frequency power supply 31 determines the j -th temporal change pattern of the source frequency $f_{\text{sub.S}}$ in the waveform cycle CY which most suppresses the degree of reflection of the source radio frequency power HF, based on the $(j-1)$ -th reference pattern. The $(j-1)$ -th reference pattern is a $(j-1)$ -th temporal change pattern or a $(j-1)$ -th derived temporal pattern of a source frequency $f_{\text{sub.S}}$ generated from the $(j-1)$ -th temporal change pattern.

[0069] In the j -th optimization process, the radio frequency power supply 31 uses a temporal change pattern of a source frequency $f_{\text{sub.S}}$ including the respective source frequencies $f_{\text{sub.S}}$ of a plurality of j -th time points (phases) in each waveform cycle CY and an interpolated source frequency $f_{\text{sub.S}}$ in each of a plurality of j -th division periods $DP_{\text{sub.j}}$, in the time series.

[0070] The number of the plurality of j -th time points in each waveform cycle CY is greater than the number of the plurality of j -th time points in each waveform cycle CY in the $(j-1)$ -th optimization process. The waveform cycle CY is divided into the plurality of j -th division periods $DP_{\text{sub.j}}$ by a plurality of j -th time points. The number of the plurality of j -th division periods $DP_{\text{sub.j}}$ in each waveform cycle CY is greater than the number of the plurality of $(j-1)$ -th division periods $DP_{\text{sub.(j-1)}}$ in the waveform cycle CY in the $(j-1)$ -th optimization process. The source frequency $f_{\text{sub.S}}$ at the initial j -th time point $T_{\text{sub.j1}}$ and the source frequency $f_{\text{sub.S}}$ at the final time point TH in the waveform cycle CY are the same.

[0071] In the example shown in FIG. 7, the plurality of third time points include a time point $T_{\text{sub.31}}$ to a time point $T_{\text{sub.3b}}$. In the example shown in FIG. 7, the plurality of third division periods $DP_{\text{sub.3}}$ include a period between the time point $T_{\text{sub.31}}$ and the time point $T_{\text{sub.32}}$, a period between the time point $T_{\text{sub.32}}$ and the time point $T_{\text{sub.33}}$, a period between the time point $T_{\text{sub.33}}$ and the time point $T_{\text{sub.34}}$, a period between the time point $T_{\text{sub.34}}$ and the time point $T_{\text{sub.35}}$, a period between the time point $T_{\text{sub.35}}$ and the time point $T_{\text{sub.36}}$, a period between the time point $T_{\text{sub.36}}$ and the time point $T_{\text{sub.37}}$, a period between the time point $T_{\text{sub.37}}$ and the time point $T_{\text{sub.38}}$, a period between the time point $T_{\text{sub.38}}$ and the time point $T_{\text{sub.39}}$, a period between the time point $T_{\text{sub.39}}$ and the time point $T_{\text{sub.3a}}$, a period between the time point $T_{\text{sub.3a}}$ and the time point $T_{\text{sub.3b}}$, and a period between the time point $T_{\text{sub.3b}}$ and the final time point $T_{\text{sub.E}}$ in the waveform cycle CY. The source frequency $f_{\text{sub.S}}$ at the initial third time point $T_{\text{sub.31}}$ and the source frequency $f_{\text{sub.S}}$ at the final time point $T_{\text{sub.E}}$ in the waveform cycle CY are the same.

[0072] A plurality of j -th time points (phases) in each waveform cycle CY may be predetermined. Alternatively, the plurality of j -th time points may be set such that each of the plurality of j -th division periods $DP_{\text{sub.j}}$ has the j -th change amount as the change amount of the load impedance of the radio frequency power supply 31 with respect to the $(j-1)$ -th reference pattern described above. In this case, the j -th change amount is smaller than the $(j-1)$ -th change amount.

Alternatively, the plurality of j -th time points may be set such that each of the plurality of j -th division periods $DP.sub.j$ has the j -th change amount as the change amount in the source frequency with respect to the $(j-1)$ -th reference pattern. Even in this case, the j -th change amount is smaller than the $(j-1)$ -th change amount. Alternatively, the plurality of j -th time points may be set such that a section in the waveform cycle CY is divided into a greater number of division periods as the load impedance of the radio frequency power supply **31** or the degree of reflection of the source radio frequency power HF is greater in the section. Alternatively, in a case where the electric bias EB includes the voltage pulse VP , the plurality of j -th time points may be set such that a section in the waveform cycle CY is divided into a greater number of division periods as the change amount in the voltage in the substrate support **11** according to the voltage pulse VP is greater in the section. The plurality of j -th time points may include a plurality of time points that are the same as the plurality of $(j-1)$ -th time points in the waveform cycle CY , respectively.

[0073] In the j -th optimization process, the radio frequency power supply **31** changes the temporal change pattern of the source frequency $f.sub.S$ of the waveform cycle $CY.sub.m$ in the time series from a temporal change pattern of the source frequency $f.sub.S$ in a preceding waveform cycle $CY.sub.m-Mj$. In the initial stage of the j -th optimization process, the temporal change pattern of the source frequency $f.sub.S$ of the waveform cycle CY is obtained by changing the $(j-1)$ -th reference pattern.

[0074] Specifically, the radio frequency power supply **31** sequentially changes the source frequency $f.sub.S$ at each of the plurality of j -th time points in the time series of the plurality of waveform cycles CY . More specifically, in the j -th optimization process, the radio frequency power supply **31** sets the source frequency $f.sub.S[m, T.sub.jn]$ at the j -th time point $T.sub.jn$ in the waveform cycle $CY.sub.m$ in the time series to a frequency having one change with respect to the source frequency $f.sub.S[m-M.sub.j, T.sub.jn]$ at the j -th time point $T.sub.jn$ in the preceding waveform cycle $CY.sub.m-Mj$. The one change is a decrease or an increase in frequency. “ m ” is a natural number representing an order of the waveform cycle CY in the time series. “ n ” is a natural number representing an order of each of the plurality of j -th time points in the waveform cycle CY . “ Mj ” is an integer which is equal to or greater than 1, and is, for example, 1. In addition, $f.sub.S[m, T.sub.jn]$ represents the source frequency $f.sub.S$ at the j -th time point $T.sub.jn$ in the waveform cycle $CY.sub.m$.

[0075] The radio frequency power supply **31** interpolates the source frequency $f.sub.S$ of each of the plurality of j -th division periods $DP.sub.j$ from the source frequency $f.sub.S$ at the immediately before time point and the source frequency $f.sub.S$ at the immediately after time point among the plurality of j -th time points and the time point $T.sub.E$. The radio frequency power supply **31** can use linear interpolation in the interpolation of the source frequency $f.sub.S$. Alternatively, the radio frequency power supply **31** may use interpolation by a high-order expression in the interpolation of the source frequency $f.sub.S$.

[0076] In the j -th optimization process, in a case where the degree of reflection of the source radio frequency power HF is reduced due to the one change, the radio frequency power supply **31** sets the source frequency $f.sub.S[m+Mj, T.sub.jn]$ in the subsequent waveform cycle $CY.sub.m+Mj$ in the time series to a frequency having the one change with respect to the source frequency $f.sub.S[m, T.sub.jn]$. In a case where the degree of reflection of the source radio frequency power HF is increased due to the one change, the radio frequency power supply **31** sets the frequency $f.sub.S[m+Mj, T.sub.jn]$ in the subsequent waveform cycle $CY.sub.m+Mj$ in the time series to a frequency having the other change with respect to the source frequency $f.sub.S[m, T.sub.jn]$. The other change is a decrease or an increase in frequency, which is a change opposite to the one change.

[0077] The radio frequency power supply **31** may sequentially optimize the respective source frequencies $f.sub.S$ of the plurality of j -th time points to suppress the degree of reflection of the source radio frequency power HF to determine the j -th temporal change pattern in the j -th

optimization process. Alternatively, the radio frequency power supply **31** may determine the j -th temporal change pattern by repeating the sequential adjustment (or change) of the source frequency $f_{\text{sub.S}}$ of each of the plurality of j -th time points to determine the j -th temporal change pattern. In any case, the order of the time point at which the source frequency $f_{\text{sub.S}}$ is optimized or adjusted among the plurality of j -th time points may be predetermined, or may be any order. For example, the source frequency $f_{\text{sub.S}}$ may be optimized or adjusted sequentially from a time point at which it is empirically known that the change in the source frequency $f_{\text{sub.S}}$ at that time point has a large influence on the degree of reflection among the plurality of j -th time points.

[0078] In the j -th optimization process, the radio frequency power supply **31** determines, as the j -th temporal change pattern, the temporal change pattern of the source frequency $f_{\text{sub.S}}$ of the waveform cycle CY in a case where the degree of reflection of the source radio frequency power HF satisfies the j -th condition. The j -th condition is predetermined as a condition indicating that the degree of reflection of the source radio frequency power HF is sufficiently suppressed. For example, the j -th condition may be satisfied in a case where the degree of reflection of the source radio frequency power HF is included in a j -th allowable range. In FIG. 7, the third reference pattern is shown by a solid line, and the temporal change pattern used in the third optimization process is shown by a broken line.

[0079] The radio frequency power supply **31** may use the temporal change pattern, which is obtained by the final optimization process and most suppresses the degree of reflection in the waveform cycle CY, as the optimal temporal change pattern, and may use the temporal change pattern in the subsequent waveform cycle CY in the time series.

[0080] In one embodiment, in a case where the degree of reflection of the source radio frequency power HF becomes large and does not satisfy the predetermined condition while the j -th optimization process (j is an integer which is equal to or greater than 2) is performed, the radio frequency power supply **31** may perform the first optimization process again. After that, the radio frequency power supply **31** may further perform the second to J -th optimization processes. According to the embodiment, in a case where abnormal discharge of plasma occurs during the j -th optimization process, it is possible to search for the temporal change pattern of the source frequency $f_{\text{sub.S}}$ that suppresses the degree of reflection of the source radio frequency power HF within the waveform cycle CY again.

[0081] In one embodiment, the radio frequency power supply **31** may obtain the $(j-1)$ -th derived temporal pattern of the source frequency $f_{\text{sub.S}}$ from the $(j-1)$ -th temporal change pattern after the $(j-1)$ -th optimization process and before the j -th optimization process.

[0082] Specifically, the radio frequency power supply **31** shifts the $(j-1)$ -th temporal change pattern in the temporal direction in a plurality of waveform cycles CY in the time series. The minimum amount of shift in the temporal direction may be smaller than a time pitch of a plurality of discrete time points at which the source frequency is set in the waveform cycle CY. The radio frequency power supply **31** may determine, as the $(j-1)$ -th derived temporal pattern, a temporal change pattern which most suppresses the degree of reflection of the source radio frequency power HF in the waveform cycle CY among the plurality of temporal change patterns obtained by shifting the $(j-1)$ -th temporal change pattern in the temporal direction. The $(j-1)$ -th derived temporal pattern can be used as a $(j-1)$ -th reference pattern in the j -th optimization process.

[0083] In the example shown in FIG. 8, the radio frequency power supply **31** obtains the derived temporal pattern of the source frequency $f_{\text{sub.S}}$ from the first temporal change pattern after the first optimization process and before the second optimization process. The radio frequency power supply **31** shifts the first temporal change pattern in the temporal direction in a plurality of waveform cycles CY in the time series. The radio frequency power supply **31** may determine, as the first derived temporal pattern, a temporal change pattern which most suppresses the degree of reflection of the source radio frequency power HF among the plurality of temporal change patterns obtained by shifting the first temporal change pattern in the temporal direction. The first derived

temporal pattern can be used as a first reference pattern in the second optimization process. In FIG. 8, a solid line indicates the first temporal change pattern, and a broken line indicates the plurality of temporal change patterns obtained by shifting the first temporal change pattern in the temporal direction.

[0084] The above-described process of determining the temporal change pattern for suppressing reflection by shifting the temporal change pattern of the source frequency $f_{\text{sub.S}}$ in the waveform cycle CY in the temporal direction may be used at any timing in each of the first to J-th optimization processes.

[0085] In one embodiment, the source radio frequency power HF and the electric bias EB are supplied simultaneously and continuously. That is, in one embodiment, a continuous wave of the source radio frequency power HF and a continuous wave of the electric bias EB are simultaneously supplied. In a case where the continuous wave of the source radio frequency power HF and the continuous wave of the electric bias EB are simultaneously supplied, the first process, which is one type of the above-described optimization process, is applied. In the first process, the above-described optimization process is applied to a plurality of consecutive waveform cycles CY as the plurality of waveform cycles CY in the time series.

[0086] In another embodiment, the bias power supply 32 periodically supplies the pulse EBP of the electric bias EB to the bias electrode. The pulse EBP is supplied to the bias electrode in each of a plurality of pulse periods PP. Each of the plurality of pulse periods PP includes repetition of the waveform cycle CY. That is, in each of the plurality of pulse periods PP, the electric bias EB is periodically supplied to the bias electrode.

[0087] Here, FIG. 9A, FIG. 9B, FIG. 10A, and FIG. 10B are referred to. Each of FIG. 9A, FIG. 9B, FIG. 10A, and FIG. 10B is a timing chart of an example of the source radio frequency power HF and the electric bias EB. In each drawing, “ON” of the source radio frequency power HF indicates that the source radio frequency power HF is supplied, and “OFF” of the source radio frequency power HF indicates that the supply of the source radio frequency power HF is stopped. In addition, in the drawings, “ON” of the electric bias EB indicates that the electric bias EB is applied to the bias electrode, and “OFF” of the electric bias EB indicates that the electric bias EB is not applied to the bias electrode. In addition, in the drawings, “HIGH” of the electric bias EB indicates that the electric bias EB having a level higher than the level of the electric bias EB indicated by “LOW” is applied to the bias electrode.

[0088] The plurality of pulse periods PP appear in temporal sequence. The plurality of pulse periods PP sequentially appear at time intervals (periods) that are the reciprocal of the pulse frequency. In the following description, a pulse period PPK represents a k-th pulse period among the plurality of pulse periods PP. The pulse frequency is lower than the bias frequency, and is, for example, a frequency which is equal to or higher than 1 kHz and equal to or lower than 100 kHz. As described above, the electric bias EB is periodically applied to the bias electrode in each of the plurality of pulse periods PP. In a period other than the plurality of pulse periods PP, the electric bias EB may not be applied to the bias electrode. Alternatively, the electric bias EB having a level lower than the level of the electric bias EB in the plurality of pulse periods PP may be applied to the bias electrode in a period other than the plurality of pulse periods PP.

[0089] As shown in FIG. 9A, the source radio frequency power HF may be supplied as a continuous wave. In the example shown in FIG. 9A, a plurality of overlapping periods OP in which the source radio frequency power HF and the electric bias EB are simultaneously supplied match the plurality of pulse periods PP, respectively.

[0090] Alternatively, as shown in FIG. 9B, FIG. 10A, and FIG. 10B, a pulse of the source radio frequency power HF may be supplied. As shown in FIG. 9B, the pulse of the source radio frequency power HF may be supplied in each of the plurality of periods that respectively match a plurality of pulse periods PP. In the example shown in FIG. 9B, the plurality of overlapping periods OP in which the source radio frequency power HF and the electric bias EB are simultaneously

supplied match the plurality of pulse periods PP, respectively. As shown in FIG. 10A and FIG. 10B, the pulse of the source radio frequency power HF may be supplied in each of the plurality of periods partially overlapping with the plurality of pulse periods PP. In the example shown in each of FIG. 10A and FIG. 10B, each of the plurality of overlapping periods OP in which the source radio frequency power HF and the electric bias EB are simultaneously supplied is a part of the corresponding pulse period PP among the plurality of pulse periods PP. In the following description, an overlapping period OP.sub.k represents a k-th overlapping period among the plurality of overlapping periods OP. In addition, a waveform cycle CY.sub.m represents an m-th waveform cycle among the plurality of waveform cycles CY in each of the plurality of overlapping periods OP.

[0091] Hereinafter, the optimization process in a case where the bias power supply 32 periodically supplies the pulse EBP of the electric bias EB to the bias electrode will be described with reference to FIG. 11A to FIG. 11C. Each of FIG. 11A to FIG. 11C is a timing chart of an example of the electric bias.

[0092] The radio frequency power supply 31 uses an initial temporal pattern prepared in advance as a temporal pattern of the source frequency f.sub.S of each of the waveform cycles CY.sub.1 to CY.sub.Ma in each of the overlapping periods OP.sub.1 to OP.sub.Ka. Ka is an integer which is equal to or greater than 1, and is, for example, 5. In addition, Ma is an integer which is equal to or greater than 1.

[0093] The radio frequency power supply 31 performs the first process on the waveform cycle CY.sub.Ma+1 to the waveform cycle CY.sub.M in each of the overlapping periods OP.sub.1 to OP.sub.Ka as the time series of the plurality of consecutive waveform cycles CY.

[0094] In the overlapping period OP.sub.Ka+1 to the final overlapping period, the radio frequency power supply 31 applies the second process to each of a plurality of time series. Each of the plurality of time series includes a waveform cycle CY.sub.m of the same order among the waveform cycle CY.sub.1 to the waveform cycle CY.sub.Mb in each of the overlapping periods OP.sub.Ka+1 to the final overlapping period. That is, the plurality of time series include the first to Mb-th time series. Mb is an integer which is equal to or greater than 1. Each of the plurality of time series includes a plurality of waveform cycles CY. Specifically, the first time series includes the waveform cycle CY.sub.1 in each of the overlapping periods OP.sub.Ka+1 to the final overlapping period. The second time series includes the waveform cycle CY.sub.2 in each of the overlapping periods OP.sub.Ka+1 to the final overlapping period. The Mb-th time series includes the waveform cycle CY.sub.Mb in each of the overlapping periods OP.sub.Ka+1 to the final overlapping period. In the second process, each of the plurality of time series is the time series of the plurality of waveform cycles CY to which the above-described optimization process (that is, the first optimization process to the J-th optimization process) is applied.

[0095] In addition, the radio frequency power supply 31 applies the first process described above to the waveform cycle CY.sub.Mb+1 to the waveform cycle CY.sub.M in each of the overlapping period OP.sub.Ka+1 to the final overlapping period as the time series of the plurality of consecutive waveform cycles CY.

[0096] According to the plasma processing apparatus 1 described above, it is possible to reduce the degree of reflection of the source radio frequency power HF in the waveform cycle CY. In addition, according to the plasma processing apparatus 1, it is possible to smoothly change the source frequency f.sub.S in each waveform cycle CY to suppress the degree of reflection of the source radio frequency power HF.

[0097] Hereinafter, a frequency control method according to one example embodiment will be described with reference to FIGS. 12 to 14. FIG. 12 is a flowchart of a frequency control method according to one example embodiment. Each of FIGS. 13 and 14 is a flowchart illustrating an example of an operation STA of the frequency control method according to one example embodiment. A frequency control method shown in FIG. 12 (hereinafter, referred to as a “method

MT”) may be performed using the plasma processing apparatus **1**. In the method MT, each unit of the plasma processing apparatus **1** may be controlled by the controller **2**.

[0098] The method MT starts in operation STA. In operation STA, the source radio frequency power HF is supplied from the radio frequency power supply **31** to generate the plasma from the gas in the chamber **10**. In operation STB, the electric bias EB is supplied to the bias electrode of the substrate support **11**. Step STA is performed when operation STB is being performed.

[0099] As shown in FIGS. **13** and **14**, in operation STa of operation STA, the above-described initial temporal pattern is used as the temporal pattern of the source frequency $f_{\text{sub.S}}$ in one or more initial waveform cycles CY in the time series of the plurality of waveform cycles CY.

[0100] Next, in operation STb to operation STc of operation STA, the first optimization process is performed. In the first optimization process, as described above, the respective source frequencies $f_{\text{sub.S}}$ at the plurality of first time points and the interpolated source frequency in each of the plurality of first division periods DP.sub.1 are used, while the respective source frequency $f_{\text{sub.S}}$ at the plurality of first time points are changed in the time series of the plurality of waveform cycles CY.

[0101] In operation STb, as described above for the first optimization process, the temporal change pattern of the source frequency $f_{\text{sub.S}}$ of the waveform cycle CY.sub.m in the time series is changed from the temporal change pattern of the source frequency $f_{\text{sub.S}}$ in the preceding waveform cycle CY.sub.m-M1.

[0102] In subsequent operation STJa, it is determined whether or not the degree of reflection of the source radio frequency power HF satisfies the first condition. In a case where the first condition is not satisfied, operation STb is performed again. In a case where the first condition is satisfied, the temporal change pattern finally obtained in the first optimization process is determined as the first temporal change pattern in operation STc.

[0103] The method MT may include a operation STd performed after the first optimization process and before the second optimization process. In operation STd, as described above, the temporal change pattern which most suppresses the degree of reflection of the source radio frequency power HF is determined as the first derived temporal pattern from among the plurality of temporal change patterns obtained by shifting the first temporal change pattern in the temporal direction.

[0104] In operation STe to operation STf of the method MT, the above-described second optimization process is performed. In the second optimization process, as described above, the respective source frequencies $f_{\text{sub.S}}$ at the plurality of second time points and the interpolated source frequency in each of the plurality of second division periods DP.sub.2 are used, while the respective source frequency $f_{\text{sub.S}}$ at the plurality of second time points are changed in the time series of the plurality of waveform cycles CY.

[0105] In operation STe, as described above for the second optimization process, the temporal change pattern of the source frequency $f_{\text{sub.S}}$ of the waveform cycle CY.sub.m in the time series is changed from the temporal change pattern of the source frequency $f_{\text{sub.S}}$ in the preceding waveform cycle CY.sub.m-M2. In the initial stage of the second optimization process, the temporal change pattern of the source frequency $f_{\text{sub.S}}$ of the waveform cycle CY is obtained by changing the first reference pattern.

[0106] After operation STe, operation STJb may be performed. In operation STJb, it is determined whether or not the degree of reflection of the source radio frequency power HF becomes large as the predetermined condition is not satisfied by using the changed temporal change pattern of the source frequency $f_{\text{sub.S}}$ in the waveform cycle CY.sub.m. As a result of the determination in operation STJb, in a case where it is determined that the degree of reflection of the source radio frequency power HF becomes large as the predetermined condition is not satisfied by using the changed temporal change pattern of the source frequency $f_{\text{sub.S}}$ in the waveform cycle CY.sub.m, the process returns to operation STa. On the other hand, as a result of the determination in operation STJb, in a case where the degree of reflection of the source radio frequency power HF

does not become large as the predetermined condition is not satisfied by using the changed temporal change pattern of the source frequency $f_{\text{sub.S}}$ in the waveform cycle $CY_{\text{sub.m}}$, the process proceeds to operation STJc.

[0107] In the subsequent operation STJc, it is determined whether or not the degree of reflection of the source radio frequency power HF satisfies the second condition described above. In a case where the second condition is not satisfied, operation STe is performed again. In a case where the second condition is satisfied, the temporal change pattern finally obtained in the second optimization process is determined as the above-described second temporal change pattern in operation STf.

[0108] The method MT may further include operation STg to operation STh. In operation STg to operation STh, the third optimization process is performed. In the third optimization process, as described above, the respective source frequencies $f_{\text{sub.S}}$ at the plurality of third time points and the interpolated source frequency in each of the plurality of third division periods $DP_{\text{sub.3}}$ are used, while the respective source frequency $f_{\text{sub.S}}$ at each of a plurality of third time points are changed in the time series.

[0109] In operation STg, as described above for the third optimization process, the temporal change pattern of the source frequency $f_{\text{sub.S}}$ of the waveform cycle $CY_{\text{sub.m}}$ in the time series is changed from the temporal change pattern of the source frequency $f_{\text{sub.S}}$ in the preceding waveform cycle $CY_{\text{sub.m-M3}}$. In the initial stage of the third optimization process, the temporal change pattern of the source frequency $f_{\text{sub.S}}$ of the waveform cycle CY is obtained by changing the second reference pattern.

[0110] After operation STg, operation STJd may be performed. In operation STJd, it is determined whether or not the degree of reflection of the source radio frequency power HF becomes large as the predetermined condition is not satisfied by using the changed temporal change pattern of the source frequency $f_{\text{sub.S}}$ in the waveform cycle $CY_{\text{sub.m}}$. As a result of the determination of operation STJd, in a case where it is determined that the degree of reflection of the source radio frequency power HF becomes large as the predetermined condition is not satisfied by using the changed temporal change pattern of the source frequency $f_{\text{sub.S}}$ in the waveform cycle $CY_{\text{sub.m}}$, the process returns to operation STa. On the other hand, as a result of the determination in operation STJd, in a case where the degree of reflection of the source radio frequency power HF does not become large as the predetermined condition is not satisfied by using the changed temporal change pattern of the source frequency $f_{\text{sub.S}}$ in the waveform cycle $CY_{\text{sub.m}}$, the process proceeds to operation STJe. In subsequent operation STJe, it is determined whether or not the degree of reflection of the source radio frequency power HF satisfies a third condition. In a case where the third condition is not satisfied, operation STg is performed again. In a case where the third condition is satisfied, the temporal change pattern finally obtained in the third optimization process is determined as the third temporal change pattern in operation STh.

[0111] The method MT may be completed after operation STh.

[0112] Alternatively, in the method MT, the fourth to J-th optimization processes may be further performed as described above. For the fourth to J-th optimization processes, reference is made to the description of the j-th optimization process described above.

[0113] While various example embodiments have been described above, various additions, omissions, substitutions and changes may be made without being limited to the example embodiments described above. Elements of the different embodiments may be combined to form another embodiment.

[0114] In another embodiment, the plasma processing apparatus may be an inductively coupled plasma processing apparatus, an ECR plasma processing apparatus, a helical wave excitation plasma processing apparatus, or a surface wave plasma processing apparatus. In any plasma processing apparatus, source radio frequency power HF is used for generating plasma.

[0115] Here, various example embodiments included in the present disclosure are described in the

following [E1] to [E16]

[E1]

[0116] A plasma processing apparatus comprising: [0117] a chamber; [0118] a substrate support disposed in the chamber; [0119] a radio frequency power supply configured to supply a source radio frequency power to generate plasma from a gas in the chamber; and [0120] a bias power supply configured to supply an electric bias having a waveform cycle to the substrate support, [0121] wherein the radio frequency power supply is configured to [0122] determine a first temporal change pattern of a source frequency of the source radio frequency power in the waveform cycle which most suppresses a degree of reflection of the source radio frequency power, by using respective source frequencies of the source radio frequency power at a plurality of first time points in the waveform cycle and an interpolated source frequency in each of a plurality of first division periods of the waveform cycle divided by the plurality of first time points, while changing the respective source frequency at the plurality of first time points, and [0123] determine a second temporal change pattern of the source frequency in the waveform cycle which most suppresses a degree of reflection of the source radio frequency power, by using the respective source frequencies of the source radio frequency power at a plurality of second time points in the waveform cycle and an interpolated source frequency in each of a plurality of second division periods of the waveform cycle divided by the plurality of second time points, while changing the respective source frequency at the plurality of second time points based on the first temporal change pattern or a temporal change pattern of the source frequency generated from the first temporal change pattern, and [0124] a number of the plurality of second division periods is greater than a number of the plurality of first division periods.

[E2]

[0125] The plasma processing apparatus according to E1, [0126] wherein the radio frequency power supply is configured to [0127] sequentially optimize the respective source frequencies at the plurality of first time points to suppress the degree of reflection of the source radio frequency power to determine the first temporal change pattern, and [0128] sequentially optimize the respective source frequencies at the plurality of second time points to suppress the degree of reflection of the source radio frequency power to determine the second temporal change pattern.

[E3]

[0129] The plasma processing apparatus according to E1, [0130] wherein the radio frequency power supply is configured to [0131] repeat sequentially adjusting the respective source frequencies of the plurality of first time points to determine the first temporal change pattern, and [0132] repeat sequentially adjusting the respective source frequencies of the plurality of second time points to determine the second temporal change pattern.

[E4]

[0133] The plasma processing apparatus according to any one of E1 to E3, [0134] wherein the plurality of first time points and the plurality of second time points are predetermined.

[E5]

[0135] The plasma processing apparatus according to any one of E1 to E3, [0136] wherein the plurality of first time points are set such that each of the plurality of first division periods has a first change amount as a change amount of a load impedance of the radio frequency power supply with respect to a temporal change pattern of the source frequency in the waveform cycle that is initially used to determine the first temporal change pattern, and [0137] the plurality of second time points are set such that each of the plurality of second division periods has a second change amount as a change amount of a load impedance of the radio frequency power supply with respect to the first temporal change pattern or the temporal change pattern of the source frequency generated from the first temporal change pattern.

[E6]

[0138] The plasma processing apparatus according to E5, [0139] wherein the second change

amount is smaller than the first change amount.

[E7]

[0140] The plasma processing apparatus according to any one of E1 to E3, [0141] wherein the plurality of first time points are set such that each of the plurality of first division periods has a first change amount as a change amount of the source frequency with respect to a temporal change pattern of the source frequency in the waveform cycle that is initially used to determine the first temporal change pattern, and [0142] the plurality of second time points are set such that each of the plurality of second division periods has a second change amount as a change amount of the source frequency with respect to the first temporal change pattern or the temporal change pattern created from the first temporal change pattern.

[E8]

[0143] The plasma processing apparatus according to E7, [0144] wherein the second change amount is smaller than the first change amount.

[E9]

[0145] The plasma processing apparatus according to any one of E1 to E3, [0146] wherein the plurality of first time points and the plurality of second time points are set such that a section in the waveform cycle is divided into a greater number of division periods as a load impedance of the radio frequency power supply or the degree of reflection of the source radio frequency power in the section is greater.

[E10]

[0147] The plasma processing apparatus according to any one of E1 to E9, [0148] wherein the electric bias is a bias radio frequency power having the waveform cycle, or the electric bias includes a voltage pulse generated at a time interval equal to a time length of the waveform cycle.

[E11]

[0149] The plasma processing apparatus according to any one of E1 to E3, [0150] wherein the electric bias includes a voltage pulse generated at a time interval equal to a time length of the waveform cycle, and [0151] the plurality of first time points and the plurality of second time points are set such that a section within the waveform cycle is divided into a greater number of division periods as a change amount in a voltage in the substrate support according to the voltage pulse in the section is greater.

[E12]

[0152] The plasma processing apparatus according to any one of E1 to E4, [0153] wherein the electric bias includes a voltage pulse generated at a time interval equal to a time length of the waveform cycle, and [0154] one of the plurality of first division periods includes a period during which the voltage pulse is supplied to the substrate support.

[E13]

[0155] The plasma processing apparatus according to any one of E1 to E12, [0156] wherein the radio frequency power supply is configured to [0157] determine a temporal change pattern of the source frequency in the waveform cycle which most suppresses the degree of reflection of the source radio frequency power, by using a temporal change pattern obtained by shifting the first temporal change pattern in a temporal direction while changing an amount of shift in the temporal direction, and [0158] use the determined temporal change pattern as the temporal change pattern of the source frequency generated from the first temporal change pattern.

[E14]

[0159] The plasma processing apparatus according to any one of E1 to E13, [0160] wherein the radio frequency power supply is configured to perform re-obtaining of the first temporal change pattern in a case where the degree of reflection of the source radio frequency power becomes so large that a predetermined reference is not satisfied when the second temporal change pattern is determined.

[E15]

[0161] A power supply system comprising: [0162] a radio frequency power supply configured to generate a source radio frequency power used to generate plasma; and [0163] a bias power supply configured to generate an electric bias used to attract ions from the plasma, the electric bias having a waveform cycle, [0164] wherein the radio frequency power supply is configured to [0165] determine a first temporal change pattern of a source frequency of the source radio frequency power in the waveform cycle which most suppresses a degree of reflection of the source radio frequency power, by using respective source frequencies of the source radio frequency power at a plurality of first time points in the waveform cycle and an interpolated source frequency in each of a plurality of first division periods of the waveform cycle divided by the plurality of first time points, while changing the respective source frequency at the plurality of first time points, and [0166] determine a second temporal change pattern of the source frequency in the waveform cycle which most suppresses a degree of reflection of the source radio frequency power, by using the respective source frequencies of the source radio frequency power at a plurality of second time points in the waveform cycle and an interpolated source frequency in each of a plurality of second division periods of the waveform cycle divided by the plurality of second time points, while changing the respective source frequency at the plurality of second time points based on the first temporal change pattern or a temporal change pattern of the source frequency generated from the first temporal change pattern, and [0167] a number of the plurality of second division periods is greater than a number of the plurality of first division periods.

[E16]

[0168] A frequency control method comprising: [0169] supplying a source radio frequency power from a radio frequency power supply to generate plasma from gas in a chamber of a plasma processing apparatus; and [0170] supplying an electric bias having a waveform cycle to a substrate support disposed in the chamber, [0171] wherein the supplying a source radio frequency power includes [0172] determining a first temporal change pattern of a source frequency of the source radio frequency power in the waveform cycle which most suppresses a degree of reflection of the source radio frequency power, by using respective source frequencies of the source radio frequency power at a plurality of first time points in the waveform cycle and an interpolated source frequency in each of a plurality of first division periods of the waveform cycle divided by the plurality of first time points, while changing the respective source frequency at the plurality of first time points, and [0173] determining a second temporal change pattern of the source frequency in the waveform cycle which most suppresses a degree of reflection of the source radio frequency power, by using the respective source frequencies of the source radio frequency power at a plurality of second time points in the waveform cycle and an interpolated source frequency in each of a plurality of second division periods of the waveform cycle divided by the plurality of second time points, while changing the respective source frequency at the plurality of second time points based on the first temporal change pattern or a temporal change pattern of the source frequency generated from the first temporal change pattern, and [0174] a number of the plurality of second division periods is greater than a number of the plurality of first division periods.

[0175] From the foregoing description, it will be appreciated that various embodiments of the present disclosure have been described herein for purposes of illustration, and that various modifications may be made without departing from the scope and spirit of the present disclosure. Accordingly, the various embodiments disclosed herein are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Claims

1. A plasma processing apparatus comprising: a chamber; a substrate support disposed in the chamber; a radio frequency power supply configured to supply a source radio frequency power to generate plasma from a gas in the chamber; and a bias power supply configured to supply an

electric bias having a waveform cycle to the substrate support, wherein the radio frequency power supply is configured to determine a first temporal change pattern of a source frequency of the source radio frequency power in the waveform cycle which most suppresses a degree of reflection of the source radio frequency power, by using respective source frequencies of the source radio frequency power at a plurality of first time points in the waveform cycle and an interpolated source frequency in each of a plurality of first division periods of the waveform cycle divided by the plurality of first time points, while changing the respective source frequency at the plurality of first time points, and determine a second temporal change pattern of the source frequency in the waveform cycle which most suppresses a degree of reflection of the source radio frequency power, by using the respective source frequencies of the source radio frequency power at a plurality of second time points in the waveform cycle and an interpolated source frequency in each of a plurality of second division periods of the waveform cycle divided by the plurality of second time points, while changing the respective source frequency at the plurality of second time points based on the first temporal change pattern or a temporal change pattern of the source frequency generated from the first temporal change pattern, and a number of the plurality of second division periods is greater than a number of the plurality of first division periods.

2. The plasma processing apparatus according to claim 1, wherein the radio frequency power supply is configured to sequentially optimize the respective source frequencies at the plurality of first time points to suppress the degree of reflection of the source radio frequency power to determine the first temporal change pattern, and sequentially optimize the respective source frequencies at the plurality of second time points to suppress the degree of reflection of the source radio frequency power to determine the second temporal change pattern.

3. The plasma processing apparatus according to claim 1, wherein the radio frequency power supply is configured to repeat sequentially adjusting the respective source frequencies of the plurality of first time points to determine the first temporal change pattern, and repeat sequentially adjusting the respective source frequencies of the plurality of second time points to determine the second temporal change pattern.

4. The plasma processing apparatus according to claim 1, wherein the plurality of first time points and the plurality of second time points are predetermined.

5. The plasma processing apparatus according to claim 1, wherein the plurality of first time points are set such that each of the plurality of first division periods has a first change amount as a change amount of a load impedance of the radio frequency power supply with respect to a temporal change pattern of the source frequency in the waveform cycle that is initially used to determine the first temporal change pattern, and the plurality of second time points are set such that each of the plurality of second division periods has a second change amount as a change amount of a load impedance of the radio frequency power supply with respect to the first temporal change pattern or the temporal change pattern of the source frequency generated from the first temporal change pattern.

6. The plasma processing apparatus according to claim 5, wherein the second change amount is smaller than the first change amount.

7. The plasma processing apparatus according to claim 1, wherein the plurality of first time points are set such that each of the plurality of first division periods has a first change amount as a change amount of the source frequency with respect to a temporal change pattern of the source frequency in the waveform cycle that is initially used to determine the first temporal change pattern, and the plurality of second time points are set such that each of the plurality of second division periods has a second change amount as a change amount of the source frequency with respect to the first temporal change pattern or the temporal change pattern created from the first temporal change pattern.

8. The plasma processing apparatus according to claim 7, wherein the second change amount is smaller than the first change amount.

- 9.** The plasma processing apparatus according to claim 1, wherein the plurality of first time points and the plurality of second time points are set such that a section in the waveform cycle is divided into a greater number of division periods as a load impedance of the radio frequency power supply or the degree of reflection of the source radio frequency power in the section is greater.
- 10.** The plasma processing apparatus according to claim 1, wherein the electric bias is a bias radio frequency power having the waveform cycle, or the electric bias includes a voltage pulse generated at a time interval equal to a time length of the waveform cycle.
- 11.** The plasma processing apparatus according to claim 1, wherein the electric bias includes a voltage pulse generated at a time interval equal to a time length of the waveform cycle, and the plurality of first time points and the plurality of second time points are set such that a section within the waveform cycle is divided into a greater number of division periods as a change amount in a voltage in the substrate support according to the voltage pulse in the section is greater.
- 12.** The plasma processing apparatus according to claim 1, wherein the electric bias includes a voltage pulse generated at a time interval equal to a time length of the waveform cycle, and one of the plurality of first division periods includes a period during which the voltage pulse is supplied to the substrate support.
- 13.** The plasma processing apparatus according to claim 1, wherein the radio frequency power supply is configured to determine a temporal change pattern of the source frequency in the waveform cycle which most suppresses the degree of reflection of the source radio frequency power, by using a temporal change pattern obtained by shifting the first temporal change pattern in a temporal direction while changing an amount of shift in the temporal direction, and use the determined temporal change pattern as the temporal change pattern of the source frequency generated from the first temporal change pattern.
- 14.** The plasma processing apparatus according to claim 1, wherein the radio frequency power supply is configured to perform re-obtaining of the first temporal change pattern in a case where the degree of reflection of the source radio frequency power becomes so large that a predetermined reference is not satisfied when the second temporal change pattern is determined.
- 15.** A power supply system comprising: a radio frequency power supply configured to generate a source radio frequency power used to generate plasma; and a bias power supply configured to generate an electric bias used to attract ions from the plasma, the electric bias having a waveform cycle, wherein the radio frequency power supply is configured to determine a first temporal change pattern of a source frequency of the source radio frequency power in the waveform cycle which most suppresses a degree of reflection of the source radio frequency power, by using respective source frequencies of the source radio frequency power at a plurality of first time points in the waveform cycle and an interpolated source frequency in each of a plurality of first division periods of the waveform cycle divided by the plurality of first time points, while changing the respective source frequency at the plurality of first time points, and determine a second temporal change pattern of the source frequency in the waveform cycle which most suppresses a degree of reflection of the source radio frequency power, by using the respective source frequencies of the source radio frequency power at a plurality of second time points in the waveform cycle and an interpolated source frequency in each of a plurality of second division periods of the waveform cycle divided by the plurality of second time points, while changing the respective source frequency at the plurality of second time points based on the first temporal change pattern or a temporal change pattern of the source frequency generated from the first temporal change pattern, and a number of the plurality of second division periods is greater than a number of the plurality of first division periods.
- 16.** A frequency control method comprising: supplying a source radio frequency power from a radio frequency power supply to generate plasma from gas in a chamber of a plasma processing apparatus; and supplying an electric bias having a waveform cycle to a substrate support disposed in the chamber, wherein the supplying a source radio frequency power includes determining a first temporal change pattern of a source frequency of the source radio frequency power in the

waveform cycle which most suppresses a degree of reflection of the source radio frequency power, by using respective source frequencies of the source radio frequency power at a plurality of first time points in the waveform cycle and an interpolated source frequency in each of a plurality of first division periods of the waveform cycle divided by the plurality of first time points, while changing the respective source frequency at the plurality of first time points, and determining a second temporal change pattern of the source frequency in the waveform cycle which most suppresses a degree of reflection of the source radio frequency power, by using the respective source frequencies of the source radio frequency power at a plurality of second time points in the waveform cycle and an interpolated source frequency in each of a plurality of second division periods of the waveform cycle divided by the plurality of second time points, while changing the respective source frequency at the plurality of second time points based on the first temporal change pattern or a temporal change pattern of the source frequency generated from the first temporal change pattern, and a number of the plurality of second division periods is greater than a number of the plurality of first division periods.
